

EXERCISES

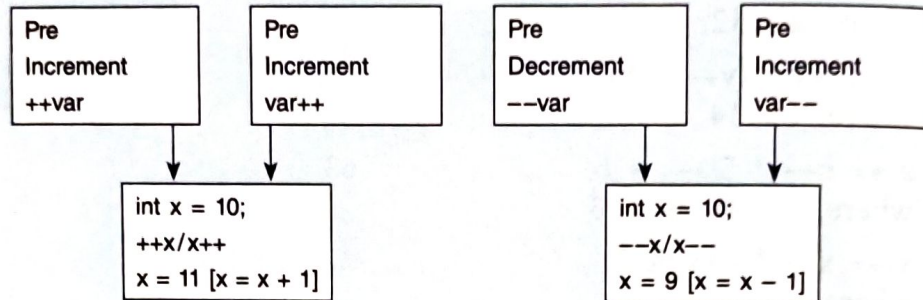


PART A – COMPETENCY FOCUSED QUESTIONS

I. Multiple Choice Questions

Tick (✓) the correct answer:

1.



Out of the following options, which operator is shown in this illustration?

- (a) Unary Operator
 - (b) Arithmetical Operator
 - (c) Ternary Operator
 - (d) Binary Operator
2. The statement `n += 4` is equivalent to:
- (a) `++n`
 - (b) `n=n+4`
 - (c) `n+1`
 - (d) none of these
3. What will be the output of a and b in the following?
if `int a, b; a=10; b=a++;`
- (a) 10,10
 - (b) 10,11
 - (c) 11,10
 - (d) 11,11
4. What will be the output of `a++` when `int a = -1`?
- (a) 1
 - (b) -1
 - (c) 0
 - (d) none of these
5. If `int a = 25, b = 5, c = 0;` what value will be stored in c, when `c = a%b`?
- (a) 5.0
 - (b) 5
 - (c) 0
 - (d) none of these
6. What will be the value of m when the given statement is executed?
`int m=8; m*=8;`
- (a) 8
 - (b) 64
 - (c) 16
 - (d) 88
7. `double c; int x, y, z; x = 5; y = 10; z = 11;`
`c = x*y+z/2;`
What will be the value stored in c?
- (a) 55.0
 - (b) 55.5
 - (c) 55
 - (d) none of these
8. `int m, p; m = 5; p = 0; p = m -- + -- m;` The value of p will be:
- (a) 11
 - (b) 10
 - (c) 8
 - (d) 12
9. `int a=7, p=0, q=0; p= + + a + - - a; q - = p;` The value of q will be:
- (a) 13
 - (b) 14
 - (c) 15
 - (d) -15
10. Identify the type of operator `& &`:
- (a) Ternary
 - (b) Unary
 - (c) Logical
 - (d) Relational
11. Which of the following logical operators is a unary operator?
- (a) `||`
 - (b) `!`
 - (c) `&&`
 - (d) None of these

[ICSE 2023]

12. Read the following text and choose the correct answer:

These operators define when the increment or decrement operators are written after the operand. They increase or decrease the value after the action has taken place.

Which of the following operators describe the above statement?

- (a) Infix (b) Out fix (c) Prefix (d) Postfix

II. Write the Java expressions for the following:

1. $a^2 + bc$

2. $\frac{a^2 - b^2}{ab}$

3. $ut + \frac{1}{2}at^2$

4. $\frac{uv}{u + v}$

5. $(a + b)^2 + b$ [ICSE-2009]

6. $2(lb+bh+lh)$

7. $a^2 + b^2$ [ICSE-2006]

8. $x^3 + y^3 - \frac{xy}{3}$ [ICSE-2006]

III. Assertion and Reasoning based questions:

Assertion (A): Operators are the special signs or symbols used to perform operations in Java programming.

Reason (R): They are considered as tokens.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
(c) Assertion (A) is true and Reason (R) is false.
(d) Assertion (A) is false and Reason (R) is true.

IV. Application based questions:

Java language also uses some special operators for dynamic allocation of the memory and implementing classes of a package. The operators can also be used to write a long expression in short form. Normally, we use BEDMAS system of hierarchical evaluation of the arithmetical expressions.

Base on the above discussion, answer the following questions:

- (a) Which operator is used for dynamic allocation of main memory?
(i) this (ii) new (iii) postfix (iv) assignment
(b) Which of the following have the same precedence of operators?
(i) + and * (ii) & and && (iii) * and / (iv) < and <<
(c) _____ operator facilitates importing Java packages in a program.
(i) .(dot) (ii) @ (iii) # (iv) &
(d) What will be the precedence of logical operators?
(i) &&, || and ! (ii) ||, && and !
(iii) !, && and || (iv) !, || and &&

V. Unsolved Java Programs:

1. Write a program to find and display the value of the given expressions:

(a) $\frac{x + 3}{6} - \frac{2x + 5}{3}$; taking the value of $x = 5$

(b) $\frac{a^2 + b^2 + c^2}{abc}$; taking the values $a=5, b=4, c=3$

2. A person is paid ₹350 for each day he works and fined ₹30 for each day he remains absent. Write a program to calculate and display his monthly income, if he is present for 25 days and remains absent for 5 days.
3. In a competitive examination, there were 150 questions. One candidate got 80% correct and the other candidate 72% correct. Write a program to calculate and display the correct answers each candidate got.
4. Write a program to find and display the percentage difference, when:
 - (a) a number is updated from 80 to 90
 - (b) a number is updated from 7.5 to 7.2
5. The normal temperature of human body is 98.6°F. Write a program to convert the temperature into degree Celsius and display the output.
 (Hint: $\frac{c}{5} = \frac{f - 32}{9}$)
6. The angles of a quadrilateral are in the ratio 3:4:5:6. Write a program to find and display all of its angles. [Hint: The sum of angles of a quadrilateral = 360°]

PART B – SUBJECTIVE QUESTIONS

I. Answer the following questions:

1. What is an operator?
2. Name the different types of operators.
3. Explain the following:
 - (a) Arithmetical operator
 - (b) Relational operator
 - (c) Logical operator
 - (d) Unary operator
 - (e) New operator
 - (f) Dot operator
4. What is a Ternary operator? Explain with the help of an example. [ICSE-2009]
5. Differentiate between the following:
 - (a) Binary operator and Ternary operator
 - (b) Logical AND (&&) and Logical OR(| |)
 - (c) Prefix operator and Postfix operator
 - (d) System.out.print() and System.out.println()
6. Differentiate between an operator and an expression. [ICSE-2005]
7. If m=5 and n=2 then what will be the values of m and n after execution of the following statements?
 - (a) m = n;
 - (b) n = m + m/n;
8. State the difference between = and ==. [ICSE-2005]
[ICSE-2007]
9. What will be the output for the following program segment?

```
int a=0, b=10, c=40;
a = --b + c++ + b;
System.out.println(" a = " + a);
```

[ICSE-2006]

10. What will be the output of the following if $x = 5$?
(a) $5 * ++x$; (b) $5 * x ++$; [ICSE-2005]

11. Evaluate the following expressions, if the values of the variables are:

$a = 2$, $b = 3$, and $c = 9$

(a) $a - (b++) * (-c)$; (b) $a * (++b) \% c$; [ICSE-2007]

12. If $a = 5$, $b = 9$, calculate the value of:

$a += a ++ - ++b + a$; [ICSE-2008]

13. State the output of the program snippet given below:

```
int a = 10, b = 12;
```

```
if(a >= 10)
```

```
    a++;
```

```
else
```

```
    ++b;
```

```
System.out.println(" a = " + a + " and b = " + b);
```

14. Rewrite the following using ternary operator.

(a) if($\text{income} \leq 100000$)

$\text{tax} = 0$;

else

$\text{tax} = (0.1 * \text{income})$;

(b) if($p > 5000$)

$d = p * 5 / 100$;

else

$d = p * 2 / 100$;